

Jeremías **Likerman**

Adjunct Researcher - CONICET · Associate Professor - University of Buenos Aires

Institute of Andean Studies ``Don Pablo Groeber" (IDEAN), CONICET--UBA

(+54) 230-455-7758 | jlikerman@uba.ar | Google Scholar | ORCID 0000-0003-3844-2085

Profile

Research profile: Geologist specializing in structural geology, geomechanics, and multiscale numerical modelling, with more than 15 years of experience in research, university teaching, fieldwork, and student training. His research integrates two- and three-dimensional models to study geodynamic, geothermal, and geomechanical processes in the Andes and in energy reservoirs, including subduction zones, crustal deformation, volcanic systems with geothermal potential, and natural and induced fracturing, with emphasis on the Vaca Muerta Formation.

Teaching, knowledge transfer, and management: Associate Professor at the University of Buenos Aires and Adjunct Researcher at CONICET. His university teaching career began in 2013 as a First-Class Teaching Assistant, continued as Head Teaching Assistant from 2018 to 2026, and has progressed to Associate Professor since 2026. He has taught Structural Geology, Geotectonics, Field Tectonics, Introduction to Geology, Geological Surveying, Geostatistics, and Geoinformatics; supervises and co-supervises researchers and students; coordinates national and international projects; and participates in knowledge-transfer, public science communication, and institutional management activities at IDEAN.

Academic and Postdoctoral Training

Postdoctoral Fellowship

Argentina / Spain

CONICET and Universitat Politècnica de Catalunya — *2015–2017*

- Topic: thermal evolution of oceanic lithosphere. Supervisors: Dr. Ernesto Cristallini (CONICET–UBA) and Dr. Sergio Zlotnik (UPC).

Visiting Researcher

United States

Stanford University — *2013*

- Courses in Structural Geology and MATLAB applied to Geosciences, as part of a Fulbright–Bunge and Born Foundation fellowship.

PhD in Geological Sciences

Argentina

University of Buenos Aires — *2010–2015*

- Thesis: *Prediction of fracturing on irregular geological surfaces using static methods*. Supervisor: Dr. Ernesto Cristallini.

BSc in Geological Sciences

Argentina

University of Buenos Aires — *2005–2010*

- Thesis: *Geology and structure of the Tres Cruces compound anticline, Jujuy, Argentina*. Supervisor: Dr. Ernesto Cristallini.

Appointments and Professional Experience

Current appointments

Adjunct Researcher

Argentina

CONICET–IDEAN — *2018–present*

- Research career since 2018; promoted to Adjunct Researcher in November 2023. Research lines: fracture prediction, geodynamic modelling, and numerical assessment of geothermal potential.

Associate Professor

Argentina

Department of Geological Sciences, University of Buenos Aires — *June 2026–present*

- Courses: Geostatistics, Geoinformatics, and Geological Surveying. Full-time appointment.

Institutional Coordinator

Argentina

IDEAN, CONICET–UBA — *September 2025–present*

- Member of the institute council since April 2024.

Previous teaching appointments

Head Teaching Assistant (Jefe de Trabajos Prácticos) *Argentina*

Department of Geological Sciences, University of Buenos Aires *March 2018–May 2026*

- Courses: Introduction to Geology, Geological Surveying, and Geostatistics.

First-Class Teaching Assistant (Ayudante de Primera) *Argentina*

Department of Geological Sciences, University of Buenos Aires *June 2013–September 2017*

- Courses: Structural Geology, Geotectonics, and Field Tectonics.

Courses and field activities

Geology of Unconventional Reservoirs *Argentina*

Instructor in charge *2025*

- Professional training course focused on the geological characterization of shale reservoirs.

Advanced Research Field Trip on Vaca Muerta *Argentina*

Organizer and instructor *2025*

- *Perspectives on Vaca Muerta: subsurface attributes analysed from surface exposures*.

Research visits

Visiting Researcher *Spain*

GEO3BCN–CSIC *May–June 2026*

- Thermomechanical numerical models of volcanic zones for geothermal-potential assessment. Host: Dr. Ivone Jiménez-Munt.

Visiting Researcher *Spain*

GEO3BCN–CSIC *May–September 2025*

- Thermomechanical modelling of volcanic zones for geothermal-potential assessment. Host: Dr. Ivone Jiménez-Munt.

Visiting Researcher *Spain*

GEO3BCN–CSIC *August–September 2024*

- Development of coupled numerical models for volcanic and geothermal systems.

Visiting Researcher *Spain*

BarcelonaTech (UPC) *2015, 2017, 2019, and 2022*

- Research stays at the Numerical Computation Laboratory.

Research Projects

2026	Three-dimensional models of stress transfer between adjacent Andean flat-slab segments (Principal Investigator). AECT-2026-1-0080	*Spain*
2026–2029	Analogue and numerical modelling of tectonic and structural problems related to energy resources (Researcher; PI: Ernesto O. Cristallini). UBACyT 20020250200097BA	*Argentina*
2026–2029	Tectonic-structural analysis through analogue and numerical modelling: contributions to energy resources (Researcher; PI: Daniel L. Yagupsky; grant: ARS 26,000,000). PIP 11220250100817CO	*Argentina*
2025	Geothermal potential of volcanic zones (Principal Investigator). AECT-2025-1-0009	*Spain*
2024–2025	Structural controls in volcanic systems: magmatic intrusions and damage zones related to hydrocarbons (Researcher; PI: Ernesto O. Cristallini). Williams Foundation	*Argentina*
2024–2026	Geothermal-potential estimation in volcanic zones through thermomechanical numerical models (Argentine partner lead; Spanish PI: Ivone Jiménez-Munt). LINCGL2024	*Spain*

2024	Geodynamic modelling of subduction zones and mantle plumes (Principal Investigator). AECT-2024-2-0027	*Spain*
2024–2027	Geodynamics of the Adria microplate, from mantle to surface (Researcher; PI: Ivone Jiménez-Munt). PID2022-139943NB-I00	*Spain*
2022	Geodynamic modelling of subduction zones (Principal Investigator). AECT-2022-3-0023	*Spain*
2020	Landscape dynamics and geological hazards associated with dams in mountain regions (Co-Principal Investigator). PICT-2019-04011	*Argentina*
2020	Analogue and numerical modelling: contributions to tectonic and structural problems (Co-Principal Investigator). PICT-2019-00997	*Argentina*
2018	Natural fracturing and first-order structures: predictive models applied to Cerro Domuyo (Co-Principal Investigator). UBACyT	*Argentina*
2017	Geological evolution of the Andes and its economic and environmental impact (Co-Principal Investigator). PUE 22920160100051	*Argentina*
2016	Numerical modelling of the thermal evolution of oceanic lithosphere (Principal Investigator). PICT 2015-1226	*Argentina*
2016	Application of analogue and numerical models to tectonic-structural problems (Co-Principal Investigator). PICT-2016-1407	*Argentina*

Research Supervision

Researcher supervision

Plotek, Berenice — *Argentina*

Assistant Researcher — *Since August 2025*

- **Supervisor:** Likerman, J. **Co-supervisor:** Cristallini, E.

Postdoctoral supervision

Plotek, Berenice — *Argentina*

Fault-propagation folds: a mechanical approach using finite-element numerical modelling — *2023–2025*

- **Supervision:** Cristallini, E. and Likerman, J.

Doctoral thesis supervision

Correa Luna, Clara — *Argentina*

Natural fracturing in the Vaca Muerta Formation: characterization and modelling — *2023*

- **Supervision:** Yagupsky, D. and Likerman, J.

Undergraduate thesis supervision

Chandía Sepúlveda, Alexander Ignacio — *Argentina*

Tectonic inversion of a metamorphic complex in the Deseado Massif: contributions from analogue modelling — *In progress*

- **Supervision:** Likerman, J. and Navarrete, C.

De Rosa, Tomás — *Argentina*

Quaternary deformation in the Fiambalá Valley, Catamarca — *In progress*

- **Supervision:** Likerman, J. and Sagripanti, L.

Blanco, Lucía *Argentina*

Assessment of the geothermal-electric potential of the Cerro Blanco Volcanic Complex, Catamarca: influence of rheology and structure through numerical models *In progress*

- **Supervision:** Likerman, J. and Lamberti, C.

Celis Santisteban, Celso Angel *Argentina*

Structural evolution of an inverted metamorphic complex in the Deseado Massif: contributions from numerical modelling *In progress*

- **Supervision:** Likerman, J. and Navarrete, C.

Ruiz, Luciana *Argentina*

Assessment of the geothermal-electric potential of the Cerro Blanco Volcanic Complex, Catamarca, through numerical modelling *2026*

- **Supervision:** Lamberti, C. and Likerman, J.

Gramajo, Pilar *Argentina*

Quaternary deformation in the Chaschuil Valley, western Sierra de Las Planchadas, Catamarca *2025*

- **Supervisor:** Sagripanti, L. **Co-supervisor:** Likerman, J.

Acosta, Luana *Argentina*

Geomorphology of the Río Grande valley between Bardas Blancas and Portezuelo del Viento, Mendoza *2023*

- **Supervision:** Winocur, D. and Likerman, J.

Mazzini, Martín *Argentina*

Geology of the eastern slope of Cerro Perito Moreno, El Bolsón, Río Negro *2022*

- **Supervision:** Likerman, J. and Tobal, J.

Villar, Cecilia *Argentina*

Undergraduate thesis in Geological Sciences (title not recorded in SIGEVA) *2021*

- **Co-supervisor:** Likerman, J. (2019–2021).

Relañez, Gastón *Argentina*

Structure and neotectonics of the Tabladitas area, Jujuy, Argentina *2019*

- **Supervision:** Yagupsky, D. and Likerman, J.

Ventisky, Esteban *Argentina*

Geology, structure, and fracturing of the Lajas Formation in the Covunco anticline, Neuquén *2017*

- **Supervision:** Likerman, J. and Tobal, J.

Correa Luna, Clara *Argentina*

Structural geology and fracturing in the Tres Cruces basin, Jujuy *2015*

- **Supervision:** Yagupsky, D. and Likerman, J.

Fellowships and Distinctions

2027 **Research Fellowship in Japan** *JSPS*
Kobe University

2025 **SZNet 2025 Chile Field Trip** *SZ4D*
National Science Foundation

2022 **Academic Mobility Fellowship among Partner Institutions** *AUIP*
Ibero-American University Association for Postgraduate Studies

2016 **National Scientific and Technical Research Council** *CONICET*
International Research Stay

2015 **National Scientific and Technical Research Council** *CONICET*
Postdoctoral Fellowship

2013 **Stanford University** *Doctoral fellowship*
Fulbright – Bunge & Born Foundation

2013 **National Scientific and Technical Research Council** *CONICET*
Doctoral Fellowship Type II

2010 **National Scientific and Technical Research Council** *CONICET*
Doctoral Fellowship Type I

Skills

Programming and reproducibility: LaTeX, Python, R, MATLAB, HTML5, Git, and reproducible scientific documentation.

Numerical modelling and HPC: Underworld2, ASPECT, GOLEM, and LaMEM; MPI, post-processing, and visualization of 2D/3D models.

Analogue modelling: experimental design, material characterization, PIV, and kinematic and mechanical analysis.

Spatial and geological analysis: QGIS, GMT, and PostGIS; geological mapping, seismic interpretation, structural analysis, and UAV-assisted field methods.

Fieldwork: more than 15 years of experience in the Andes; campaign planning, logistics, and team leadership.

Languages: Spanish native, English fluent (C2), Italian (A2), and Catalan (A2).

Peer-Reviewed Publications

1. Analogue modelling to enhance spatial reasoning and understanding of tectonic processes in geoscience education. Jara, P. P., Pinto, L., **Likerman, J.** Andean Geology *53.3 (2026)*
2. The role of compressional structures in the development of collapse calderas: Insights from analogue models. Villarroel, M. A., Browning, J., Jara, P. P., Harnett, C. E., Holohan, E. P., Marquardt, C. J., Guzman, M., **Likerman, J.** Journal of Volcanology and Geothermal Research *(2026) p. 108582*
3. Impact of large-scale structures on fracture network connectivity: Insights into the Vaca Muerta unconventional play, Neuquén basin, Argentina. Correa-Luna, C., Yagupsky, D. L., **Likerman, J.**, Barcelona, H. Tectonophysics *(2025) p. 230640*
4. Slab underthrusting is the primary control on flat slab size. Gianni, G. M., Gallo, L. C., **Likerman, J.**, Echaurren, A., Gianni, C. R., Faccena, C. Science Advances *11.28 (2025)*
5. Massive Jurassic slab break-off revealed by a multidisciplinary reappraisal of the Chon Aike silicic large igneous province. Navarrete, C., Gianni, G., Tassara, S., Zaffarana, C., **Likerman, J.**, Márquez, M., Wostbrock, J., Planavsky, N., Tardani, D., Frasette, M. P. Earth-Science Reviews *249 (2024) p. 104651*
6. Geomechanical modeling of fault-propagation folds: A comparative analysis of finite-element and the trishear kinematic model. Plotek, B., **Likerman, J.**, Cristallini, E. Journal of Structural Geology *(2024) p. 105064*
7. Ghost-arc geochemical anomaly at a spreading ridge caused by supersized flat subduction. Gianni, G. M., **Likerman, J.**, Navarrete, C. R., Gianni, C. R., Zlotnik, S. Nature communications *14.1 (2023) p. 2083*
8. Natural fracture characterization of the Los Catutos Member (Vaca Muerta Formation) in the southern sector of the Sierra de la Vaca Muerta, Neuquén Basin, Argentina. Correa-Luna, C., Yagupsky, D. L., **Likerman, J.**, Barcelona, H. Journal of South American Earth Sciences *120 (2022) p. 104081*
9. Kinematics of fault-propagation folding: Analysis of velocity fields in numerical modeling simulations. Plotek, B., Heckenbach, E., Brune, S., Cristallini, E., **Likerman, J.** Journal of Structural Geology *162 (2022) p. 104703*
10. The influence of climate on the dynamics of mountain building within the Northern Patagonian Andes. García Morabito, E., Beltrán-Triviño, A., Terrizzano, C. M., Bechis, F., **Likerman, J.**, Von Quadt, A., Ramos, V. A. Tectonics *40.2 (2021) e2020TC006374*
11. The effects of small-scale convection in the shallow lithosphere of the North Atlantic. **Likerman, J.**, Zlotnik, S., Li, C.-F. Geophysical Journal International *227.3 (2021) pp. 1512–1522*
12. Fracture network analysis of Yacoraite Formation in the Tres Cruces sub-basin, northwestern Argentina. Luna, C. C., Yagupsky, D. L., **Likerman, J.** Journal of South American Earth Sciences *(2019) p. 102446*
13. Cenozoic intraplate tectonics in Central Patagonia: Record of main Andean phases in a weak upper plate. Gianni, G. M., Echaurren, A., Folguera, A., **Likerman, J.**, Encinas, A., García, H. P. A., Dal Molin, C., Valencia, V. A. Tectonophysics *721 (2017) pp. 151–166*
14. Closure type effects on the structural pattern of an inverted extensional basin of variable width: Results from analogue models. Jara, P., **Likerman, J.**, Charrier, R., Herrera, S., Pinto, L., Villarroel, M., Winocur, D. Journal of South American Earth Sciences *(2017)*
15. Climatic and Tectonic forcing on alluvial fans in the Southern Central Andes. Terrizzano, C. M., Morabito, E. G., Christl, M., **Likerman, J.**, Tobal, J., Yamin, M., Zech, R. Quaternary science reviews *172 (2017) pp. 131–141*
16. Structural and statistical characterization of joints and multi-scale faults in an alternating sandstone and shale turbidite sequence at the Santa Susana Field Laboratory: Implications for their effects on groundwater flow and contaminant transport. Cilona, A., Aydin, A., **Likerman, J.**, Parker, B., Cherry, J. Journal of Structural Geology *85 (2016) pp. 95–114*
17. Middle to late Miocene extensional collapse of the North Patagonian Andes (41° 30′–42° S). Tobal, J. E., Folguera, A., **Likerman, J.**, Naipauer, M., Sellés, D., Boedo, F. L., Ramos, V. A., Gimenez, M. Tectonophysics *657 (2015) pp. 155–171*
18. Geodynamic context for the deposition of coarse-grained deep-water axial channel systems in the Patagonian Andes. Ghiglione, M. C., **Likerman, J.**, Barberón, V., Giambiagi, L. B., Aguirre-Urreta, B., Suarez, F. Basin Research *26.6 (2014) pp. 726–745*
19. Role of basin width variation in tectonic inversion: insight from analogue modelling and implications for the tectonic inversion of the Abanico Basin, 32°–34° S, Central Andes. Jara, P., **Likerman, J.**, Winocur, D., Ghiglione, M. C., Cristallini, E. O., Pinto, L., Charrier, R. Geological Society, London, Special Publications *399 (2014) SP399–7*
20. Along-strike structural variations in the Southern Patagonian Andes: Insights from physical modeling. **Likerman, J.**, Burlando, J. F., Cristallini, E. O., Ghiglione, M. C. Tectonophysics *590 (2013) pp. 106–120*